

Universal Plastic Shredder V2.0

Electrical Control and Power System

Final Report - Rough Draft

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[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]

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1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost \$10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand safety control, lid interlock, E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: \$3,000 (EE portion approximately \$1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., “shredder validated to X HP load with Y throughput rate”) once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.
- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.

- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within \$3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.
- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** GE 5KE182BC205B, 3 HP, 230V (low-voltage connection), 1750 RPM, FLA 7.6 A.
- **Brake resistor:** 75 Ω , 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** C-more Micro-Graphic EA1-T4CL 4-inch color touch panel.
- **Safety circuit:** Hardwired E-stop loop, lid interlock, two-hand deadman start. Motor contactor energized through safety chain.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset, lid interlock) → PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU -V) → VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) → PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.

- HMI ↔ PLC via Ethernet (H2-DM1E built-in Ethernet port).

[TODO: Insert system block diagram here. Convert one of the AutoCAD drawings or draw fresh in draw.io.]

5 Hardware Design

5.1 Power Distribution

- Main disconnect: **[TODO: specify rating]**.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

- Selected Huanyang HY02D211B-T over alternatives via decision matrix (price, documentation, reliability).
- Programmed motor parameters: PD141 = 230 V, PD142 = 7.6 A, PD143 = 4 poles, PD144 = 1750 RPM.
- Over-torque detection programmed: PD052 = 02 (fault indication on relay), PD123 = 3, PD124 = 150 (%), PD125 = 3.0 s.
- Brake resistor connected to P/Pr terminals to handle regenerative energy from rapid reversal during jam-clear cycles.

[NOTE: The over-torque relay function (PD052 = 12) did not fire the relay in our specific Huanyang revision when PD123 = 2 (continue running). Documented as a firmware-specific limitation in vfd/Overcurrent_Signal_Wiring.md. Working configuration uses PD052 = 02 (fault indication) and PD123 = 3 (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD's latched fault state between retry attempts.]

5.3 Safety Circuit

- E-stop: NC contact pair in hardwired loop. Drops motor contactor on activation.
- Lid interlock: NC switch in same hardwired safety loop.
- Two-hand deadman: both buttons must be held continuously to enable motor.
- Motor contactor: K1 main contactor energized through safety chain. PLC outputs RUN command *through* the contactor, so safety chain has hardware precedence over PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

5.4 Jam Detection Signal Chain

- VFD's FA/FB/FC relay programmed for Fault Indication (PD052 = 02).

- Wired as NO active-high pair: FC (relay common) connected to +24V through a 500 mA fast-blow fuse; FA (NO contact) connected to PLC X5 input on D2-08ND3 (Slot 2).
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads [X5] on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the redundant hardwired E-stop loop and motor contactor, rather than at the jam-signal level.
- Detailed wiring in repository: `vfd/Overcurrent_Signal_Wiring.md`.

6 Software Design

6.1 PLC Ladder Logic Architecture

- Do-more Designer stage program (similar to grafcet / SFC).
- States: IDLE, RUNNING, JAM_DETECTED, RST_PULSE, REVERSE, RESUME, LOCKOUT.
- Transitions driven by deadman input, X5 jam signal, jam counter, and reset button.

6.2 State Machine

- **IDLE** → **RUNNING** when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → **JAM_DETECTED** on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → **RST_PULSE** if jam count < 3, → **LOCKOUT** if jam count ≥ 3.
- **RST_PULSE** (500 ms) → **REVERSE** (2 s).
- **REVERSE** → **RESUME** (brief delay) → back to **RUNNING**.
- **LOCKOUT** → **IDLE** only via dedicated reset button.

6.3 Jam Counter Reset Logic

- Counter increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets to 0 if motor runs cleanly for 30 continuous seconds in the **RUNNING** state.
- Also resets if deadman is released for 30 continuous seconds (session reset).
- Whichever condition fires first wins.

6.4 Startup Mask

- For first 5 seconds after PLC power-up, X5 is ignored.
- In NO active-high wiring, this guards against transient X5 pulses during VFD initialization. Less critical than under NC wiring but kept as a defensive measure.

6.5 HMI Programming

- Status displays: Ready, Running, Jam in progress, Lockout, Fault.
- Operator inputs: Speed setpoint (currently fixed in PLC), Reset button.
- Data displayed: Motor current (read from VFD over Modbus – **[TODO: verify Modbus setup]**), Jam count this session, Total runtime.

[QUESTION: We have not finalized the HMI screen layout. Should we include mock-ups in the rough draft, or wait for the final?]

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- Documented in `electrical/Wire_Color_Standard.md`.
- All control wires labeled at both ends.
- All control wires terminated with crimped ferrules at terminal blocks.

7.2 Hardwired Safety Functions

- E-stop: Category 0 stop (immediate removal of power) per ISO 13849. NC fail-safe wiring.
- Lid interlock: NC fail-safe; opens motor contactor when lid raised.
- Two-hand control: per ISO 13851 (Type IIIA at minimum) – buttons must be pressed within 0.5 s of each other and held continuously.
- Overload: motor thermal protection in VFD plus separate motor contactor with overload relay.

[TODO: Confirm we are meeting Category 1 or higher per ISO 13849-1. Need to walk through the SIL/PL calculation or get a hand-wave from the advisor that capstone scope doesn't require formal SIL.]

7.3 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: same as above – relay returns to idle state which on FB-FC means contact opens, X5 LOW, locked out.
- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 Testing and Validation

8.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (PD124 = 1, PD125 = 5.0): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (PD124 = 150) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

8.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

8.3 Safety Validation

- E-stop response time: **[TODO: measure with scope]**.
- Lid interlock response: **[TODO: measure]**.
- Two-hand control timing: **[TODO: verify both buttons must be held continuously, simultaneous release stops within X ms]**.

9 Results

9.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with PLC over Ethernet.

9.2 Open Issues

- Vendor-specific Huanyang firmware quirk: PD052 = 12 does not fire relay in continue-running mode. Working around with PD052 = 02 + PD123 = 3.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

10 Discussion

10.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD (\$150) over ATO (\$450). Documented quirks in repository for future maintenance.
- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NC fail-safe versus NO active-high jam signal: chose NC for safety; trade-off is potential nuisance lockouts from broken wires (acceptable for shredder).

10.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on all control wires – loose strand connections caused multiple debugging hours during commissioning.

10.3 Project Management Notes

[TODO: Add a paragraph on how the team coordinated, what tools we used (GitHub, weekly meetings, etc.), and how schedule slipped or held.]

11 Future Work

- Wireless monitoring (read-only) – could expose PLC data via the H2-DM1E's built-in web server or an ESP32 bridge.
- Data logging: total runtime, jam events per shift, faults. Useful for predictive maintenance.
- Predictive jam detection: train a small model on current waveform shape just before jams to detect them earlier.
- Mechanical integration with feed hopper that auto-meters plastic into the blade.
- Higher-capacity model: scale to 5–10 HP for higher throughput.

12 Ethics and Design Considerations

See docs/Ethics_Design_Evaluation.docx for the full filled-out form. Summary:

- **Public health and safety:** Yes – multiple redundant safety interlocks, NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.

- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

13 References

- NFPA 70 (2023): National Electrical Code.
- NFPA 79 (2024): Electrical Standard for Industrial Machinery.
- ISO 13849-1:2023: Safety-related parts of control systems.
- ISO 13851:2019: Two-hand control devices.
- Huanyang HY-series VFD User Manual.
- AutomationDirect DL205 / Do-more User Manual.
- GE 5KE Frame 182T Motor Datasheet.

[TODO: Add full reference list with URLs / publication info before final.]

14 Appendices

14.1 Appendix A: Bill of Materials

[TODO: Insert latest BOM (BOM/BOM_02_20_26.xlsx) as a formatted table or summary.]

14.2 Appendix B: PLC I/O Point List

[TODO: Insert from PLC/Point List.pdf.]

14.3 Appendix C: Wiring Diagrams

[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]

14.4 Appendix D: VFD Parameter Settings

See repository: vfd/Overcurrent_Signal_Wiring.md, vfd/Brake_Resistor_Wiring.md, motor/Motor_Nameplate

14.5 Appendix E: Test Plan

See docs/Test_Plan_v1.0.md.

End of rough draft. Comments and questions welcome – please use Google Docs share for inline review (forthcoming).